

任务 2：知识图谱可视化

本实践任务要求每位同学独立完成提交

一. 批量导入项目 data 目录中的 CSV 数据

将 CSV 文件放入 neo4j-community-5.20.0\import 目录下
执行项目 data 目录中的导入数据命令（三个 txt 文件）

 loadnode.txt

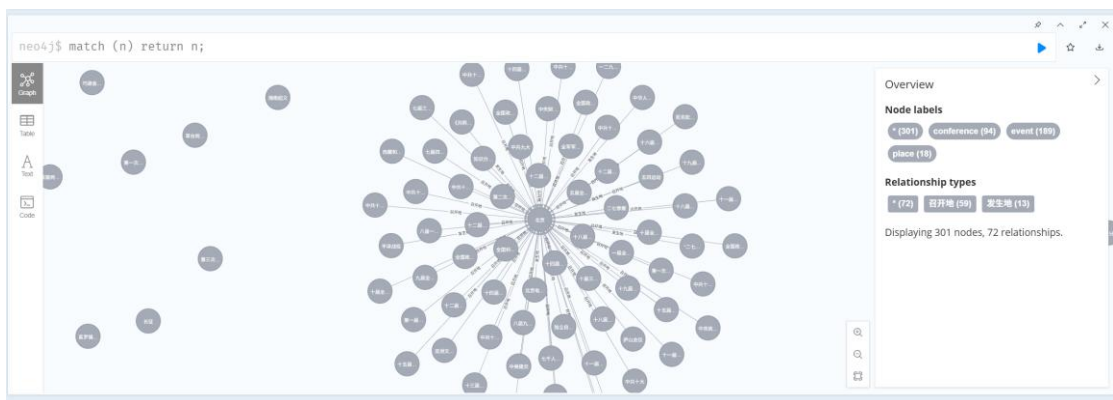
 loadrelation.txt

 loadspirit.txt

最后执行 deduplicate.txt 中的命令，进行去重复操作

 deduplicate.txt

导入数据后，通过 `match (n) return n` 查看



二. 项目执行

1. 执行前确保正确安装 nodejs 和 angular cli
2. 通过修改 server/server.js 文件来链接访问数据库

```
var express = require('express');//引用框架
var bodyParser = require('body-parser');
var app = express();//操作框架的实例对象
var username = 'neo4j'
var password = '12345678'
```

3. 分别在 server 和 view 文件夹下执行 npm install 导入 node_modules 包（这次及其之后的 npm install 需要加上—force 以覆盖依赖检查）

4. 在项目 view 目录下安装前端 AntV G6

指定 G6 版本为 4.3.3

npm install --save @antv/g6@4.3.3

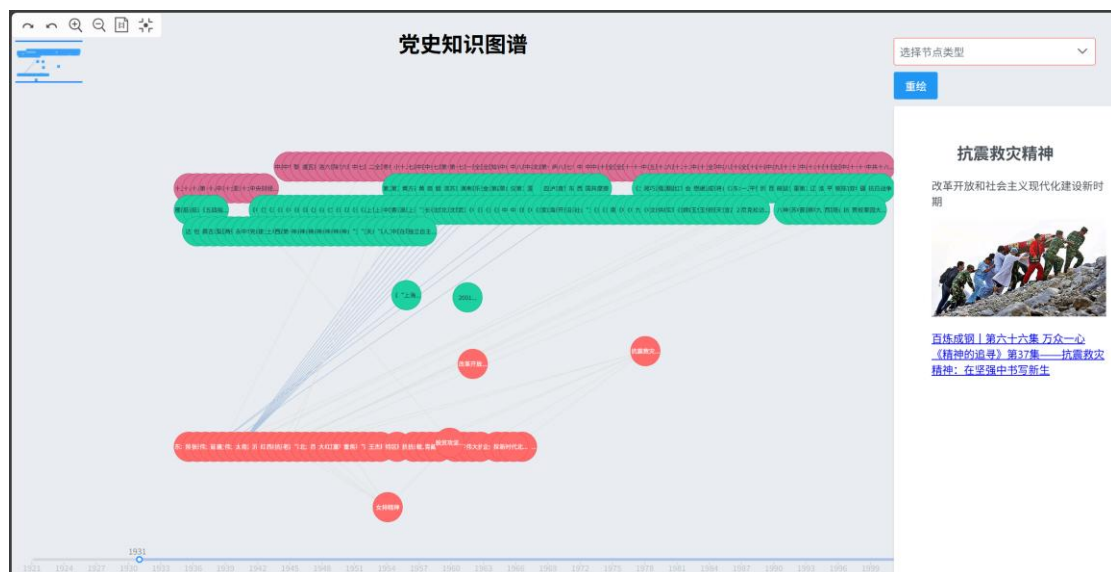
```
C:\BaiduSyncdisk\xinwang\tools\neo4j-community-5.20.0>npm install --save @antv/g6
npm WARN deprecated hulk.js@1.0.6: This package is not maintained anymore on npmjs.com, please use GitHub URL to fetch the latest version. See the package homepage for instructions.
```

added 134 packages in 15s

9 packages are looking for funding
run 'npm fund' for details

5. server 目录下执行 npm start 开启 8080 端口监听

6. view 目录下执行 npm start 打开 angular 项目



三. 代码架构

项目链接：

1. Angular 为本项目使用的连接前后端的框架，
使用文档参见：<https://angular.cn/docs>
通过 generate component + {工程名} 指令生成项目文件

g6-graph.component.css	2021/8/30 17:40	层叠样式表文档	1 KB
g6-graph.component.html	2021/8/30 17:40	Chrome HTML D...	2 KB
g6-graph.component.spec.ts	2021/8/30 17:40	TS 文件	1 KB
g6-graph.component.ts	2021/8/30 21:04	TS 文件	14 KB
interface.ts	2021/8/30 17:40	TS 文件	1 KB

2. 后端数据库存储: Neo4j 数据库

3. 前端使用 AntV G6

使用文档参考: <https://g6.antv.vision/zh/docs/manual/getting-started/>

G6 是一个图可视化引擎。它提供了图的绘制、布局、分析、交互、动画等图可视化的基础能力。旨在让关系变得透明, 简单。让用户获得关系数据的 Insight。基于 G6, 用户可以快速搭建自己的图分析或图编辑应用。

下面结合项目代码对一些实现细节进行介绍

a) 创建容器

需要在 HTML 中创建一个用于容纳 G6 绘制的图的容器, 通常为 div 标签。G6 在绘制时会在该容器下追加 canvas 标签, 然后将图绘制在其中。

g6-graph.component.html

```

1 <meta name="referrer" content="no-referrer" />
2 <div id="title" style="margin-left: 35px; margin-bottom: 10px;">知识图谱</div>
3 <div id="left" style="width: 'calc(100vw)'"><div id="container" ></div></div>
4 <div id="right" >
5   <p-dropdown styleClass="dropdown-override" [options]="types" [(ngModel)]="selectedType" optionLabel="name" [filter]="true"
6     filterBy="name" [showClear]="true" placeholder="选择节点类型" required>
7     <ng-template pTemplate="selectedItem">
8       <div class="type-item type-item-value" *ngIf="selectedType">
9         <div>{{selectedType.name}}</div>
10      </div>
11    </ng-template>
12  </p-dropdown>
13  <button pButton pRipple style="margin-bottom: 10px;" type="button" label="重绘" (click)="doGraphRender()"></button>
14  <div class="card">
15    <p-card id="card-header" [style]="{width: '100%', height: 'calc(75vh)'}" styleClass="p-card-shadow">
16      <p-scrollPanel [style]="{width: '100%', height: 'calc(65vh)'}">
17        <p id="intro">请选择节点</p>
18        <div id="outDiv"></div>
19        <p-scrollTop target="parent"></p-scrollTop>
20      </p-scrollPanel>
21    </p-card>
22  </div>
23 </div>
24
25

```

b) 数据准备

引入 G6 的数据源为 JSON 格式的对象。该对象中需要有节点 (nodes) 和边 (edges) 字段

```

ngOnInit(): void {
  this.types = [
    { name: '全部', code: 'all' },
    { name: '精神谱系', code: 'spirit' },
    { name: '人物', code: 'people' },
    { name: '会议', code: 'conference' },
    { name: '事件', code: 'event' },
    { name: '地点', code: 'place' }
  ];
  var nurl = 'http://127.0.0.1:8080/node'; //发起请求的url
  this.http.get<any>(nurl).subscribe(nres => {
    var rurl = 'http://127.0.0.1:8080/relation';
    this.http.get<any>(rurl).subscribe(rres => {
      // this.rdata = rres;
      // this.ndata = nres;
      this.createData(rres, nres);
      this.drawGraph(rres, nres);
    });
  });
}

```

c) 图实例化

图实例化时，至少需要为图设置容器、宽、高：

```

132 //
133 const graph = new G6.Graph({
134   animate: true,
135   container: 'container',
136   width: constrainBox.width,
137   height: constrainBox.height,
138   linkCenter: true,
139   fitView: true,
140   plugins: [toolBar, tooltip, minimap],
141   layout: {
142     type: 'force', //1. gForce 2.radial 3.concentric
143     preventOverlap: true,
144     // onTick,
145     linkDistance: 400,
146     unitRadius: 100,
147     //workerEnabled: true,
148     nodeStrength: -30,
149     forceSimulation: null,
150     edgeStrength: 0.7,
151   },
152   defaultEdge: {
153     style: {
154       endArrow: {

```

d) 图的渲染

数据的加载和图的渲染是两个步骤，可以分开进行。

```

169 graph.data({
170   "nodes": that.nodes,
171   "edges": that.edges
172 });
173 graph.render();
174

```

e) 元素及其配置

图的元素特指图上的节点 Node 和边 Edge。G6 内置了一系列内置的节点和内置的边，供用户自由选择。G6 不同的内置节点或不同的内置边主要区别在于元素的图形 Shape，例如，节点可以是圆形、矩形、图片等。

```

152 |
153 |
154 |
155 |     defaultEdge: {
156 |       style: {
157 |         endArrow: {
158 |           path: 'M 4,0 L -4,-4 L -4,4 Z',
159 |           d: 4
160 |         }
161 |       },
162 |       labelCfg: {
163 |         autoRotate: true,
164 |         refY: 10
165 |       }
166 |     },
167 |   },
168 | },
169 |

```

f) 使用布局 layout

当数据中没有节点位置信息，或者数据中的位置信息不满足需求时，需要借助一些布局算法对图进行布局。

本项目中，使用的是力导向布局 (Force Layout)。

➤ Force Layout: 经典力导向布局:

力导向布局: 一个布局网络中，粒子与粒子之间具有引力和斥力，从初始的随机无序的布局不断演变，逐渐趋于平衡稳定的布局方式称之为力导向布局。适用于描述事物间关系，比如人物关系、计算机网络关系等。

```

141 |
142 |
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144 |
145 |     layout: {
146 |       type: 'force', //1. gForce 2.radial 3.concentric
147 |       preventOverlap: true,
148 |       // onTick,
149 |       linkDistance: 400,
150 |       unitRadius: 100,
151 |       //workerEnabled: true,
152 |       nodeStrength: -30,
153 |       forceSimulation: null,
154 |       edgeStrength: 0.7,
155 |     },
156 |   },
157 | },
158 |

```

g) 图的交互 Behavior

G6 封装了一系列交互方法，方便用户直接使用

h) 插件与工具

为辅助用户在图上探索，G6 提供了一些辅助工具，其中一部分是插件工具，另一部分是交互工具。

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```

➤ Minimap

缩略图 (Minimap) 是一种常见的用于快速预览和探索图的工具，可作为导航辅助用户探索大规模图。



```
54 const minimap = new G6.Minimap({
55   size: [100, 100],
56   className: 'minimap',
57   type: 'delegate',
58   });
```

➤ tooltip 节点提示框



```
59 const tooltip = new G6.Tooltip({
60   offsetX: 10,
61   offsetY: 20,
62   getContent(e) {
63     const curNode = e.item.getModel();
64     const outDiv = document.createElement('div');
65     outDiv.style.width = '280px';
66     // <li>内容: ${curNode.value.content ? curNode.value.content.substring(0, 100) + "...": "无"}</li>
67     outDiv.innerHTML = `
68       <div>
69         <h4 style="text-align:center">${curNode.name}</h4>
70         <ul>
71           <li>相关链接: ${curNode.value.link1 ? "<a href='" + curNode.value.link1 + "'" + curNode.value.title1 + "</a>" : ""}
72           <li>图片: ${curNode.value.figure ? "<img src='" + curNode.value.figure + "'" + curNode.value.figure + "' style='width:240px;height:auto;'>" : ""}
73         </ul>
74       </div>
75     `;
76     return outDiv;
77   },
78   itemTypes: ['node']
79 });
```

➤ toolBar



```
52 const toolBar = new G6.ToolBar({
53   })
54 const minimap = new G6.Minimap({
55   size: [100, 100],
56   className: 'minimap',
57   type: 'delegate',
58 });
```

➤ Iconfont

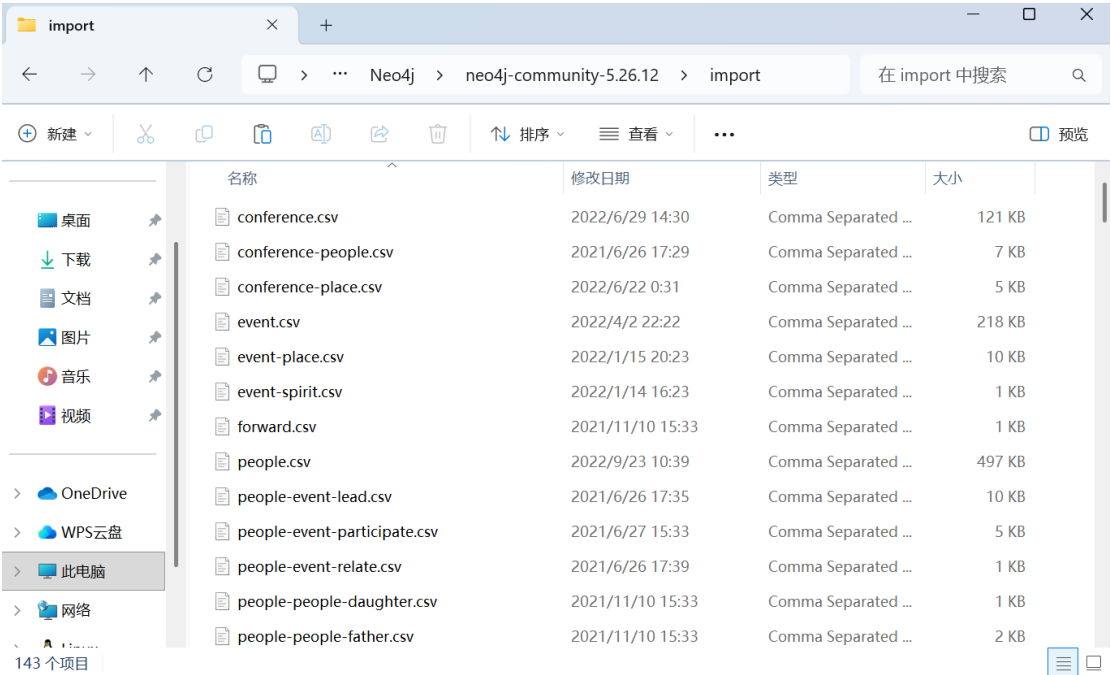
Iconfont 提供丰富的语义化矢量图形。详见阿里巴巴-iconfont 平台。

<https://ant.design/components/icon-cn/>

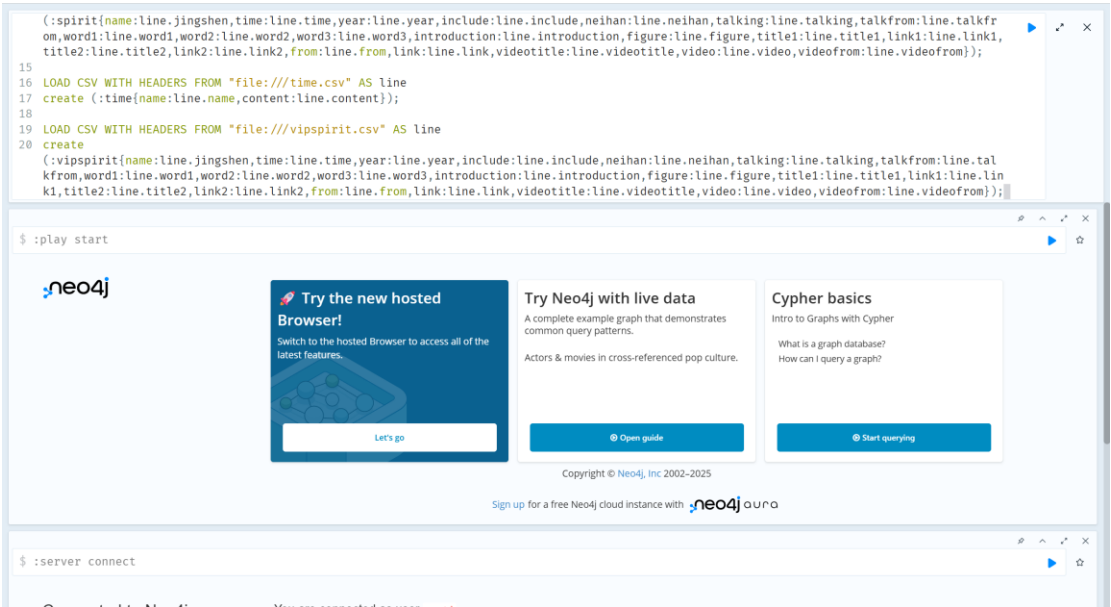
【完成任务：每个上述实践任务指导中的示例截图在自己部署的机器上重新截取一遍，按照步骤顺序将截图粘贴到下面】

一、导入数据

将 3 个文件放到指定目录下：



启动 neo4j 并输入.txt 文件中的导入数据命令：



成功执行：

```
$ LOAD CSV WITH HEADERS FROM "file:///conference.csv" AS line fieldterminator ';' create (:conference{name:line.name,year:line.year,time:line.time,...})

neo4j$ LOAD CSV WITH HEADERS FROM "file:///conference.csv" AS line fieldterminator ';' create (:conference{name:line.name,year:line.yea...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///event.csv" AS line fieldterminator ';' create (:event{name:line.name,year:line.year,link1:li...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///place.csv" AS line fieldterminator ';' create (:place{name:line.name,figure:line.figure,titl...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///people.csv" AS line fieldterminator ';' create (:people{name:line.people,birth:line.birth,de...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit.csv" AS line create (:spirit{name:line.jingshen,time:line.time,year:line.year,include...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///time.csv" AS line create (:time{name:line.name,content:line.content})
neo4j$ LOAD CSV WITH HEADERS FROM "file:///vipspirit.csv" AS line create (:vipspirit{name:line.jingshen,time:line.time,year:line.year,i...
```

然后执行 loadrelation.txt 的命令：

```
neo4j$

neo4j$ LOAD CSV WITH HEADERS FROM "file:///conference-place.csv" AS line fieldterminator ';' match (from: conference{name:line.name}),(to: place{name:li...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///event-place.csv" AS line fieldterminator ';' match (from: event{name:line.name}),(to: place...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///event-spirit.csv" AS line fieldterminator ';' match (from: spirit{name:line.jingshen}),(to:...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit.csv" AS line match (from: time{name:line.time}),(to: spirit{name:line.jingshen}) mer...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///people-event-lead.csv" AS line fieldterminator ';' match (from: people{name:line.people}),(...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///people-event-participate.csv" AS line fieldterminator ';' match (from: people{name:line.peo...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///people-event-relate.csv" AS line fieldterminator ';' match (from: people{name:line.people})...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///people-people-daughter.csv" AS line fieldterminator ';' match (from: people{name:line.peopl...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///people-people-father.csv" AS line fieldterminator ';' match (from: people{name:line.people1...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///people-people-friend.csv" AS line fieldterminator ';' match (from: people{name:line.people1...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///people-people-husband.csv" AS line fieldterminator ';' match (from: people{name:line.people...
```

执行 loadspirit.txt 的命令：

```
neo4j$

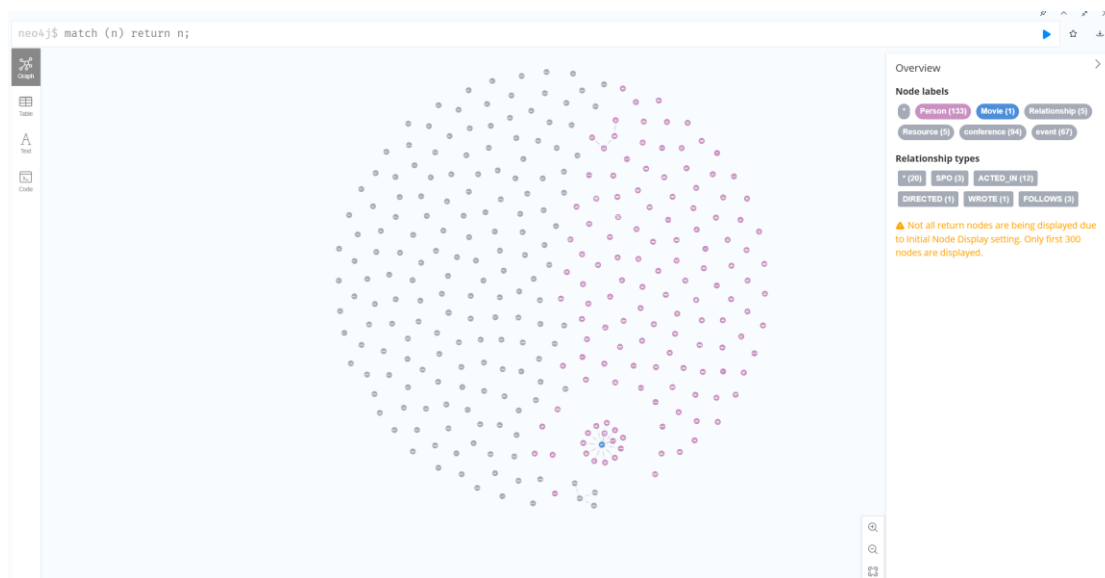
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-0.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-1.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-2.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-3.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-4.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-5.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-6.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-7.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-8.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-9.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
neo4j$ LOAD CSV WITH HEADERS FROM "file:///spirit-keyword-spirit-10.csv" AS line match (from: spirit{name:line.spirit1}),(to: spirit{name:line.spirit2}) ...
```

然后执行 deduplicate.txt 中的命令：

```
neo4j$

neo4j$ MATCH (p) WITH p ORDER BY p.name, COUNT{(p)--{}} DESC WITH p.name as name, collect(p) AS nodes WHERE size(nodes) > 1 UNWIND nodes[1..] AS n ...
Deleted 43 nodes, deleted 240 relationships, completed after 893 ms.
```

通过 match (n) return n 查看：



二. 项目执行

1.

首先安装 angular cli, 使用 node.js

```
Windows PowerShell
(base) PS E:\othersoftware\Neo4j\neo4j-community-5.26.12> npm install -g @angular/cli
(base) PS E:\othersoftware\Neo4j\neo4j-community-5.26.12> npm install -g @angular/cli

added 348 packages in 24s

66 packages are looking for funding
  run 'npm fund' for details
(base) PS E:\othersoftware\Neo4j\neo4j-community-5.26.12> ng version

Angular CLI: 20.3.2
Node: 22.19.0
Package Manager: npm 10.9.3
OS: win32 x64

Angular: <error>

Package                                Version
-----                                -
@angular-devkit/architect              0.2003.2 (cli-only)
@angular-devkit/core                   20.3.2 (cli-only)
@angular-devkit/schematics             20.3.2 (cli-only)
@schematics/angular                   20.3.2 (cli-only)
(base) PS E:\othersoftware\Neo4j\neo4j-community-5.26.12> |
```

2.修改 server/server.js 文件

```
var express = require('express');//引用框架
var bodyParser = require('body-parser');
var app = express();//操作框架的实例对象
var username = 'neo4j'
var password = 'dujunbao'
```

3.在 server 和 view 下导入 node_modules 包

```
Windows PowerShell
(base) PS E:\othersoftware\Neo4j\neo4j-community-5.26.12> .\bin\neo4j.bat status
Neo4j is running
(base) PS E:\Desktop\knowledgehijian\pal\task2> cd E:\Desktop\knowledgehijian\pal\task2
(base) PS E:\Desktop\knowledgehijian\pal\task2> npm install -force
npm WARN using --force Recommended protections disabled.
(base) PS E:\Desktop\knowledgehijian\pal\task2> cd E:\Desktop\knowledgehijian\pal\task2\code\server
(base) PS E:\Desktop\knowledgehijian\pal\task2\code\server> npm install -force
npm WARN using --force Recommended protections disabled.

added 58 packages, and audited 59 packages in 9s

8 vulnerabilities (3 low, 1 moderate, 4 high)

To address all issues, run:
  npm audit fix

Run 'npm audit' for details.
(base) PS E:\Desktop\knowledgehijian\pal\task2\code\server> cd E:\Desktop\knowledgehijian\pal\task2\code\view
(base) PS E:\Desktop\knowledgehijian\pal\task2\code\view> npm install -force
npm WARN using --force Recommended protections disabled.
npm WARN ERESOLVE overriding peer dependency
npm WARN While resolving: @angular/animations@10.2.4
npm WARN Found: @angular/core@10.2.2
npm WARN   node_modules/@angular/core
npm WARN   @angular/core@"10.2.0" from the root project
npm WARN   6 more (@angular/common, @angular/forms, ...)
npm WARN Could not resolve dependency:
npm WARN peer @angular/core@"10.2.4" from @angular/animations@10.2.4
npm WARN   node_modules/@angular/animations
npm WARN   @angular/animations@"10.2.4" from the root project
npm WARN Conflicting peer dependency: @angular/core@10.2.4
npm WARN   node_modules/@angular/core
npm WARN peer @angular/core@"10.2.4" from @angular/animations@10.2.4
npm WARN   node_modules/@angular/animations
npm WARN   @angular/animations@"10.2.4" from the root project
npm WARN ERESOLVE overriding peer dependency
npm WARN While resolving: @angular/cdk@11.1.1
npm WARN Found: @angular/common@10.2.2
npm WARN   node_modules/@angular/common
npm WARN   @angular/common@"10.2.0" from the root project
npm WARN   5 more (@angular/forms, @angular/platform-browser, ...)
npm WARN Could not resolve dependency:
npm WARN peer @angular/common@"11.0.0 || ^12.0.0-0" from @angular/cdk@11.1.1
npm WARN   node_modules/@angular/cdk
npm WARN   @angular/cdk@"11.1.1" from the root project
npm WARN   1 more (@angular/material)
npm WARN Conflicting peer dependency: @angular/common@12.2.17
npm WARN   node_modules/@angular/common
```

4.在 view 下安装前端 AntV G6

```
Windows PowerShell
To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run 'npm audit' for details.
(base) PS E:\Desktop\knowledgehijian\pal\task2\code\view> npm install --save @antv/g6@4.3.3 -force
npm WARN using --force Recommended protections disabled.
npm WARN ERESOLVE overriding peer dependency
npm WARN While resolving: @angular/animations@10.2.4
npm WARN Found: @angular/core@10.2.2
npm WARN   node_modules/@angular/core
npm WARN peer @angular/core@"10.2.2" from @angular/common@10.2.2
npm WARN   node_modules/@angular/common
npm WARN peer @angular/common@"10.2.2" from @angular/forms@10.2.2
npm WARN   node_modules/@angular/forms
npm WARN   5 more (@angular/platform-browser, ...)
npm WARN   6 more (@angular/forms, @angular/platform-browser, ...)
npm WARN Could not resolve dependency:
npm WARN peer @angular/core@"10.2.4" from @angular/animations@10.2.4
npm WARN   node_modules/@angular/animations
npm WARN   @angular/animations@"10.2.4" from the root project
npm WARN Conflicting peer dependency: @angular/core@10.2.4
npm WARN   node_modules/@angular/core
npm WARN peer @angular/core@"10.2.4" from @angular/animations@10.2.4
npm WARN   node_modules/@angular/animations
npm WARN   @angular/animations@"10.2.4" from the root project
npm WARN ERESOLVE overriding peer dependency
npm WARN While resolving: @angular/cdk@11.1.1
npm WARN Found: @angular/common@10.2.2
npm WARN   node_modules/@angular/common
npm WARN peer @angular/common@"10.2.2" from @angular/forms@10.2.2
npm WARN   node_modules/@angular/forms
npm WARN peer @angular/forms@"7.0.0 || ^8.0.0 || ^9.0.0 || ^10.0.0 || ^11.0.0" from primeng@11.4.2
npm WARN   node_modules/primeng
npm WARN   1 more (the root project)
npm WARN   5 more (@angular/platform-browser, ...)
npm WARN Could not resolve dependency:
npm WARN peer @angular/common@"11.0.0 || ^12.0.0-0" from @angular/cdk@11.1.1
npm WARN   node_modules/@angular/cdk
npm WARN peer @angular/cdk@"11.1.1" from @angular/material@11.1.1
npm WARN   node_modules/@angular/material
npm WARN   1 more (the root project)
npm WARN Conflicting peer dependency: @angular/common@12.2.17
npm WARN   node_modules/@angular/common
npm WARN peer @angular/common@"11.0.0 || ^12.0.0-0" from @angular/cdk@11.1.1
```

5.开启 8080 端口监听

```
(base) PS E:\Desktop\knowledgehijian\pal\task2\code\server> npm start

> server@1.0.0 start
> node server.js

listening on 8080
```

6. 开启项目

```
ng serve
Windows PowerShell
版权所有 (C) Microsoft Corporation. 保留所有权利。

安装最新的 PowerShell, 了解新功能和改进! https://aka.ms/PSWindows

加载个人及系统配置文件用了 5605 毫秒。
(base) PS E:\Desktop\knowledgehijian\pal\task2\code\view> npm start

> view@0.0.0 start
> set NODE_OPTIONS=--openssl-legacy-provider && ng serve

|
```

由于网络原因, 将 workerEnabled 改为 false

```
plugins: [timebar, toolBar, tooltip, minimap],
layout: {
  type: 'force', //1. gForce 2.radial 3.concentric
  preventOverlap: true,
  linkDistance: 600,
  unitRadius: 100,
  workerEnabled: false,
  nodeStrength: -50,
  forceSimulation: null,
  nodeSpacing: 5,
},
defaultEdge: {
  style: {
    endArrow: {
      path: 'M 4,0 L -4,-4 L -4,4 Z',
      d: 4
```

连接 <http://localhost:4200/>
成功显示

党史知识图谱

人物

重绘

周平丁

李长春

李长

叶笃正

生辰：1916年2月
逝世：2013年10月

叶笃正（1916.2.21-2013.10.16），又名叶平斋，出生于天津，祖籍安徽省安庆市，气象学家，中国现代气象学主要奠基人之一、中国大气物

精神谱系

精神谱系

重绘

王杰精神

社会主义革命和建设时期

[精神的追寻15王杰精神：一不怕苦、二不怕死《军事纪事》英雄团长 王杰](#)

党史知识图谱

会议

重绘

中共八大

1956年9月15日至27日

1956年9月召开的党的八大，是党在全国执政后召开的第一次全国代表大会，它以正确判断国内主要矛盾和形成领导国家发展的路线方针政策，留下了探索社会主义建设道路的历史足迹。八大的贡献是多方面